

A Change Analysis of the 2021 Mount Etna's Paroxysm Sequence

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Abstract

Mt. Etna's February-April 2021 paroxysm sequence produced measurable changes across the volcano and its surrounding landscape. This study integrates two satellite-based remote sensing approaches to characterize those changes: Land Surface Temperature (LST) derived from Landsat 7 and 8 thermal observations at 30 m resolution and 16-day revisit (accessed through Climate Engine) and Sentinel-1 Interferometric Synthetic Aperture Radar (InSAR) displacement and coherence products (accessed through the COMET Volcanic and Magmatic Deformation Portal). Monthly LST maps from December 2020 to May 2021 identify a 47-50°C thermal anomaly emerging on the southern caldera flank in February 2021, cooling through March, and dissipating by April. The InSAR cumulative displacement map shows approximately +180 mm of uplift at the caldera center and -140 to -180 mm of subsidence on the western flank, while a 2018–2022 displacement time series captures pre-eruption inflation followed by syn- and post-eruption deflation consistent with magma withdrawal. Coherence is lowest along the southern flank, spatially co-locating with the LST anomaly, and a 40 km topographic transect shows peak displacement coinciding with peak elevation at the caldera center. Both the thermal and deformation evidence confirms that the 2021 paroxysm produced localized, short-lived changes across the Etna edifice.

I. Introduction

Mt. Etna is a global anomaly in the study of volcanoes and tectonic activity. Situated on the island of Sicily, Italy (just 30km from the nearest city, Catania), humans have borne witness to the volcanic activity of Mt. Etna for over two millennia (UNESCO 2013). The eruption of Mt. Etna in 396 BCE is said to have protected the city of Catania, under the rule of Dionysius I of Syracuse, from the Carthaginian army (Britannica). The ancient Greeks spoke about the volcano through legends in 475 BCE (Britannica). The Encyclopedia Britannica records 71 documented eruptions between 1500 BCE and 1669 CE — the 1669 event being the most intense — 26 more between 1669 and 1900, and over 100 episodes since 1900. Lava-flow evidence indicates that early eruptions date back 300,000 to 500,000 years, giving Etna the longest recorded eruption history of any volcano in the world (NASA 2001; UNESCO 2013).

Not only has data historically been collected for eruptions of Mt. Etna, but the features of the volcano lend themselves well to human study (UNESCO 2013). The intensity and magnitude of Mt. Etna's eruptions make it a fascinating volcano to study. Perhaps further studies will consider comparing these events with one another or investigating potential motivators for such seismic activity. This paper focuses on the 2021 paroxysm sequence — a series of mid-intensity eruptions (Andronico et al. 2021). We ask: How did Mount Etna and the surrounding land change during the 2021 paroxysm sequence? A paroxysm sequence is a series of events, in this case “mid-intensity...eruptions” (Andronico et al. 2021). We analyze land deformation and land surface temperature to characterize the volcano's impact on the surrounding terrain.

Satellite remote sensing offers synoptic, repeatable observations of inaccessible volcanic terrain. For thermal monitoring, the Landsat program (Landsat 7 ETM+ and Landsat 8 TIRS) provides 30 m thermal-infrared imagery on a 16-day revisit cycle, from which Land Surface Temperature (LST) products can be derived to map active lava fields and track flow cooling (Marchese et al. 2021; Massimetti et al. 2020). Compared with coarser sensors such as MODIS, Landsat resolves individual flow lobes and small hotspots, and its multi-decadal archive supports comparison between active and quiescent periods.

For deformation monitoring, Interferometric Synthetic Aperture Radar (InSAR) measures ground displacement at millimeter precision from the phase difference between successive satellite passes. The European Space Agency's Sentinel-1 constellation has produced a continuous InSAR record over Etna since 2014, documenting the volcano's inflation–deflation cycles, flank-collapse dynamics, and deformation during recent paroxysms (Bruno et al. 2022; Beccaro et al. 2023; Pezzo et al. 2023). Interferogram coherence additionally flags surfaces altered between acquisitions, helping identify lava emplacement zones. This study combines Landsat-derived LST and Sentinel-1 InSAR to leverage both thermal and geodetic perspectives on the 2021 event.

II. Data Sources and Methods

Two satellite remote sensing datasets were employed: Sentinel-1 InSAR products and Landsat-derived LST. InSAR displacement, coherence, and topographic-profile products were obtained from the COMET Volcanic and Magmatic Deformation Portal (<https://comet.nerc.ac.uk/comet-volcano-portal/>), which processes Sentinel-1 C-band SAR acquisitions into pre-computed cumulative displacement maps, time series, and coherence rasters over the world's active volcanoes. LST data were accessed through Climate Engine (<https://app.climateengine.org/climateEngine>), which serves Landsat 7 ETM+ and Landsat 8 TIRS thermal observations at 30 m resolution on the native 16-day revisit cycle of each sensor.

The study began with the identification of candidate sites for analysis. Several twenty-first-century volcanic eruptions were considered, and Mount Etna was ultimately selected because of the substantial land deformation and surface temperature change associated with the 2021 paroxysm sequence. Preliminary research indicated that high-quality data were available for this event. Two parameters were subsequently identified as the focus of the investigation: land surface temperature and land deformation.

LST maps were generated in Climate Engine for the period December 2020 to June 2021. The first map (December 2020 – January 2021) served as a pre-eruption baseline, and subsequent maps were produced at one-month intervals. The temperature display range was adjusted from the default -10°C to 45°C to a range of 0°C to 50°C , since values below 0°C did not occur in the study area and distorted interpretation. Observations beyond June 2021 showed no meaningful correlation with the event and were excluded.

For InSAR, Mt. Etna was located in the COMET Deformation Portal and a displacement time series was generated for January 2019 – January 2022, with the area of interest constrained to the caldera. Two products were output: a colored displacement map and a corresponding time series graph. The analysis was extended with a coherence map (values 0–1, rendered black to white) and a topographic profile along a 40 km transect plotting displacement (mm) and elevation (m) against distance.

III. Results

Two parameters associated with the 2021 eruption were examined: change in land surface temperature throughout the event and change in land deformation before, during, and after the event.

3.1 Land Surface Temperature

The LST time series captured the spatial distribution and temporal evolution of magma flow during the event (*Figure 1*). At the 30 m resolution of Landsat thermal pixels, the values shown are mixed temperatures averaged over each pixel footprint, so a reading of $\sim 50^{\circ}\text{C}$ implies a far hotter subpixel source — fresh lava at several hundred $^{\circ}\text{C}$ occupying a small fraction of the pixel, surrounded by cooler terrain. LST therefore integrates a hot subpixel component with the background, which is why an active flow front registers as $47\text{--}50^{\circ}\text{C}$ in Landsat rather than the much higher values one might expect.



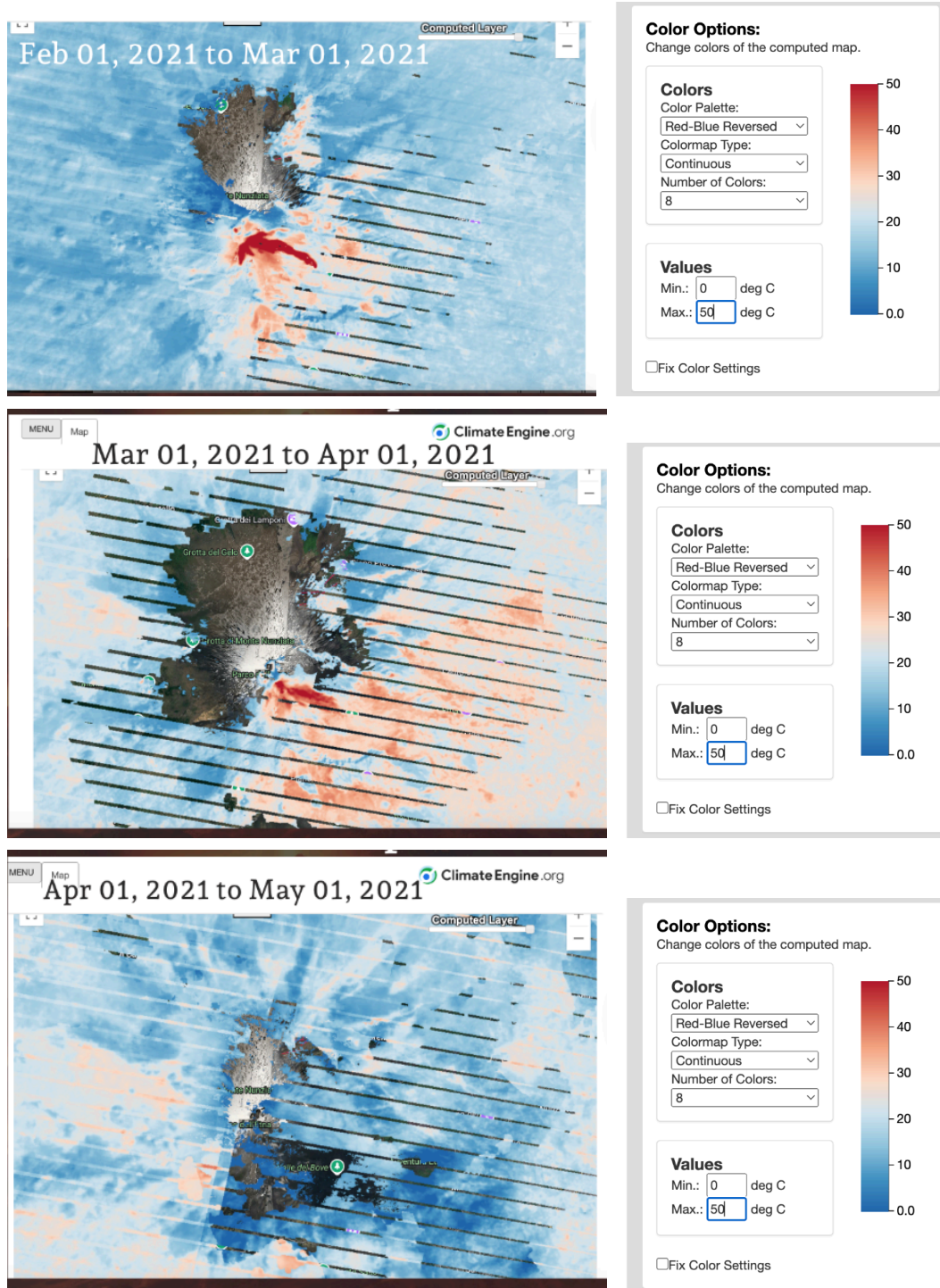


Figure 1. Monthly LST time series of Mt. Etna, December 2020 – May 2021, illustrating the thermal signature of magma flow before, during, and after the 2021 paroxysm sequence.

The first panel (December 2020 – February 2021) serves as a pre-eruption control, with generally cool surface temperatures and only a few unrelated hotspots. The second panel (February – March 2021) reveals a sharp, spatially coherent zone of elevated temperature ($\sim 47\text{--}50\text{ }^{\circ}\text{C}$) extending south-southeast from the summit along the southern flank. This is our target

magma-flow corridor for the analysis. By the third panel (March – April 2021), the corridor has cooled to roughly 40 °C with only isolated patches still at 47–50 °C, indicating that magma supply was diminishing and the deposit was cooling radiatively even before eruptive activity ceased. The fourth panel (April – May 2021) shows almost no thermal signal, marking the end of the eruptive period in agreement with the published chronology (Andronico et al. 2021; Marchese et al. 2021).

3.2 Land Deformation

The cumulative displacement map (*Figure 2*) shows positive uplift of approximately +180 mm at the center of the caldera, with the rim displaced on the order of +150 mm. The western flank exhibits substantial subsidence, ranging from approximately –140 mm to –180 mm. Localized regions on the eastern side recorded smaller positive displacements of +50 to +75 mm. Overall, the caldera center rose while the western flank sank significantly.

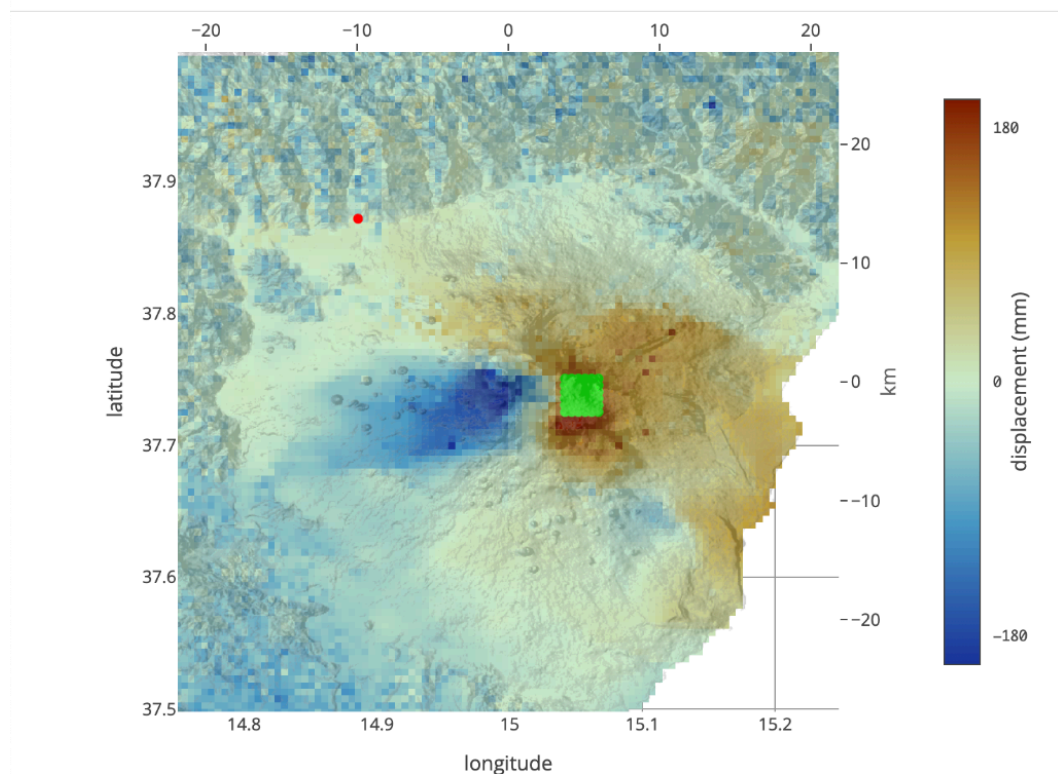


Figure 2. InSAR-derived Sentinel-1 cumulative displacement map of Mount Etna, showing positive uplift near the caldera center and pronounced subsidence along the western flank during the 2021 paroxysm sequence.

A multi-year displacement time series (*Figure 3*) provides additional temporal context. Although the principal eruptive period extended from February 2021 through April 2021, the time series provides additional context, including evidence of an earlier eruption in 2019. A pronounced positive jump in displacement is observed between 2019 and 2020, indicating renewed volcanic activity. Within the 2021 focus window, displacement increased linearly beginning in 2020, consistent with pre-eruption inflation, and subsequently exhibited a linear decline through the middle of 2021. This pattern aligns closely with the documented chronology of the event. The

observed trend can be explained by subsidence, a process whereby a volcano sinks following an eruption as more magma exits the system than enters it, with cooling further contributing to volumetric contraction (Yellowstone Volcano Observatory, 2021).

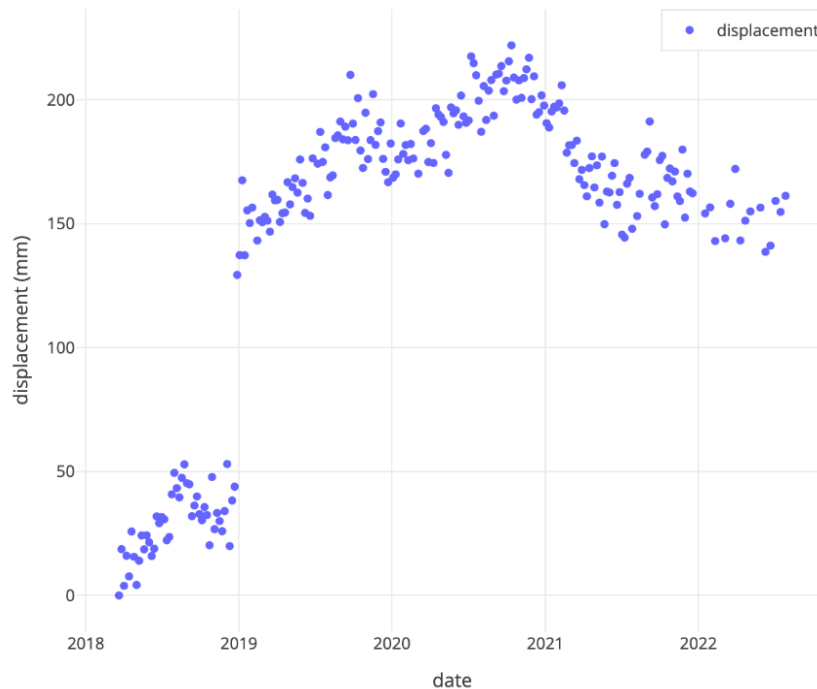


Figure 3. InSAR displacement time series at the center of Mount Etna's caldera (2018-2023), capturing the inflation phase preceding the 2021 paroxysm sequence and the subsequent deflation associated with the magma withdrawal.

The highest coherence values are located along the western edge of the caldera, while the lowest values appear at the center of the caldera and to the south (*Figure 4*). Coherence quantifies the similarity of the radar signal between two SAR acquisitions, with values ranging from 0 (complete decorrelation) to 1 (identical returns). Stable surfaces (e.g., bare rock) keep high coherence. By contrast, surfaces reworked between passes — by erosion, vegetation change, or, in this case, lava emplacement — lose coherence as the radar scatterers are rearranged. A striking spatial correspondence emerges between the magma-flow corridor identified in the LST time series and the low-coherence region beneath the green transect line. The same southern-flank zone that appeared as the 47–50 °C thermal anomaly in February 2021 (*Figure 1*) also lost coherence in the Sentinel-1 record. This indicates that the surface was transformed enough between successive acquisitions that the radar no longer recognized it as the same terrain — the geodetic fingerprint of lava emplacement, where hot material covers and reshapes the pre-existing surface.

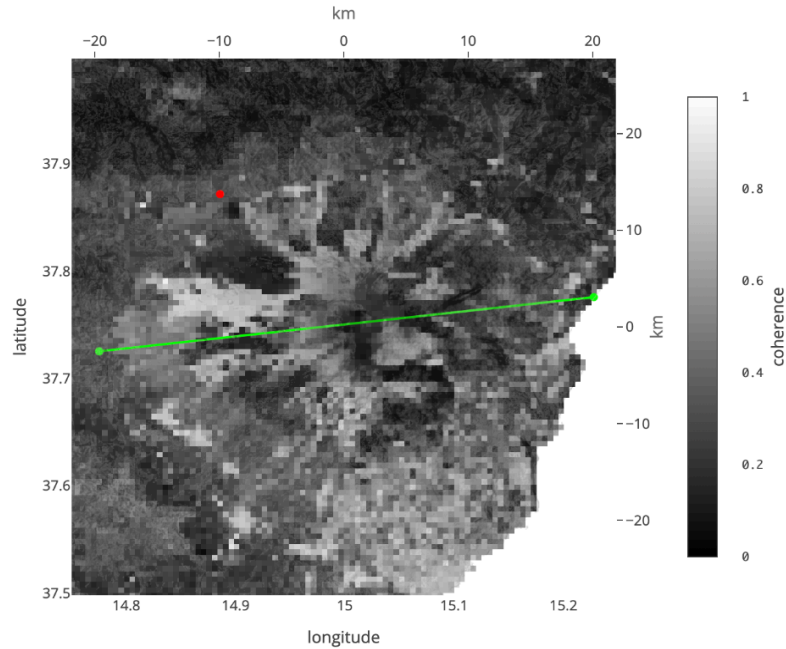


Figure 4. Sentinel-1 coherence map of Mount Etna, where low coherence values along the southern caldera correspond to the zones of greatest surface change identified in the LST and displacement analyses.

The topographic profile (Figure 5) plots displacement (mm) and elevation (m) against distance along a 40 km transect across the caldera. At 20 km there is a significant dip in negative displacement which correlates to being on the left side of the volcano which lines up with the displacement color gradient map. So, at that point elevation and displacement have zero correlation. But, as you follow along the transect, the peak of the volcano is closely linked to displacement around 25 km along the transect. At 25 km across it is where the center of the caldera is, so in this case there is higher correlation between elevation and displacement. Near the peak is also where the largest positive displacement happened.

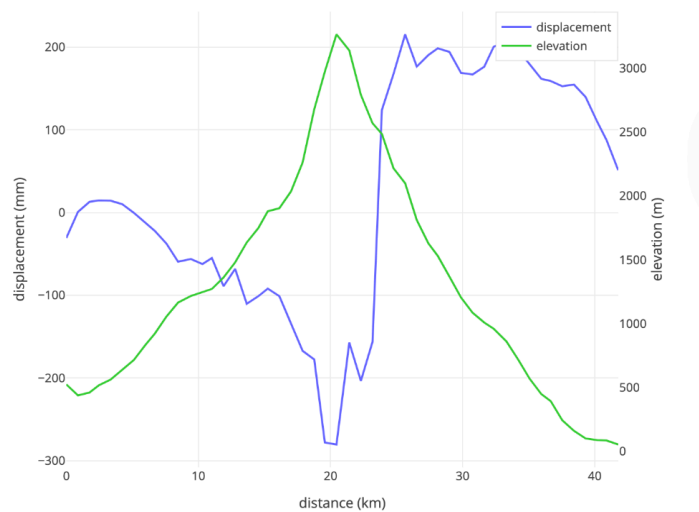


Figure 5. A topographic profile along a 40 km transect crossing Mount Etna's caldera, showing the relationship between surface displacement (mm) and elevation (m) and highlighting the coupling between peak elevation and maximum positive displacement at the caldera center.

IV. Conclusion

By integrating InSAR-derived land deformation data with Land Surface Temperature (LST) time series, this study constructed a coherent, multi-dimensional picture of the dynamic and layered processes that show the changes of the volcano and its surrounding landscape between February and April 2021.

The LST time series revealed the thermal fingerprint of magma flow: a concentrated heat signature peaking at 47–50 °C along the southern caldera flank in February, cooling through March, and largely dissipating by April. This progression aligns with the documented chronology of the paroxysm sequence, validating both our methodology and data sources. The displacement map and time series show uplift near the caldera center and subsidence on the western flank — the signature of a volcano exhaling, as magma is expelled, pressure is released, and the land sinks in response. The coherence map confirms that low-coherence values spatially overlap the zones of greatest thermal and deformation activity.

These converging lines of evidence illustrate that the 2021 paroxysm drove measurable, spatially distinct, and temporally bounded changes across the Mt. Etna system. On top of that, the human context has been particularly meaningful: Catania sits just 30km away, and the agricultural lands surrounding the volcano silently absorb the costs of each eruption.

Future work could substantially deepen this framework by incorporating NDVI-based vegetation loss assessments, GNSS ground-truth validation, and a broader temporal comparison across multiple paroxysm sequences – building toward a more complete assessment of the eruption's ecological and humanitarian footprint on the communities living in Etna's long shadow. Ultimately, this would provide an integrated early-warning system that informs evacuation planning, land-use policy, and agricultural resilience for the communities in Etna's path.

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